

Keyu Wang

☎ (+49) 1525 6272814 | ✉ keyu.wang@student.uni-tuebingen.de | 🌐 website | 🎓 Google Scholar | 🐙 GitHub | 🔗 LinkedIn

EDUCATION

University of Tuebingen

M.Sc. in Machine Learning

Tuebingen, Germany

Oct. 2024 – Present

Current GPA: 1.6 (German scale; 1.0 = best, 5.0 = worst)

Coursework: Non-convex Optimization for Deep Learning, Statistical Machine Learning, Probabilistic Machine Learning, The Science of Machine Learning Benchmarks, Understanding LLMs, etc.

Southeast University

B.Eng. in Artificial Intelligence

Nanjing, China

Sep. 2020 – Jun. 2024

Cumulative GPA: 3.79/4.0

Coursework: Machine Learning, Deep Learning, Natural Language Processing, Reinforcement Learning, etc.

SKILLS

Programming: Python (PyTorch), C/C++.

Expertise: LLM Pretraining, Supervised fine-tuning, Inference optimization, Model evaluation

Languages: Chinese (Native), English (Fluent, IELTS 7.5/C1).

WORKING EXPERIENCE

Research Assistant

Max Planck Institute for Intelligent Systems

Tuebingen, Germany

Dec. 2025 – Present

- Work on **depth-effective scaling of foundation models** to better utilize and preserve computational depth through architectural design, pre-training strategy, and model compression.
- **Architectures:** Developed DEPTHBENCH to measure how residual connection affects computational depth, by systematically varying the width–depth aspect ratio ($d_{\text{model}}/n_{\text{layer}}$) while keeping the model size and pre-training recipe fixed across 10 architectures (e.g., mHC, AttnRes); Revealed residual connection design as a key determinant of whether depth can serve as a meaningful scaling axis. 📄 [Preprint 2026](#)
- **Pre-training:** Developed an adaptive pre-training approach with layerwise learning rates guided by Heavy-Tail Theory to promote more balanced optimization across layers, faster convergence and improved generalization. 📄 [ICML 2026](#)
- **Compression:** Studied how layer pruning affects sequential and parallel test-time scaling, showing that even mild pruning can impair reasoning diversity and self-correction, while standard supervised fine-tuning only partially recovers the lost capabilities. 📄 [COLM 2026](#)

PUBLICATIONS

- [1] **Keyu Wang***, Yangyi Huang*, Jiale Kang, David González-Martínez, Weiyang Liu, and Shiwei Liu. “DepthBench: Measuring How Residual Connections Enable More Computational Depth.” **Under Review, arXiv 2026**.
- [2] Di He, Songjun Tu, **Keyu Wang**, Lu Yin, and Shiwei Liu. “One LR Doesn’t Fit All: Heavy-Tail Guided Layerwise Learning Rates for LLMs.” **ICML 2026**.
- [3] **Keyu Wang***, Tian Lyu*, Guinan Su, Lu Yin, Marco Canini, Jonas Geiping, and Shiwei Liu. “When Fewer Layers Break More Chains: Layer Pruning Harms Test-time Scaling in LLMs.” **COLM 2026**.